M.TECH COMPUTER SCIENCE & ENGINEERING
CAPSTONE RESEARCH & IMPLEMENTATION PROJECT

Implementation of Quantum Phase Estimation (QPE) Algorithm for Different Cases

A Rigorous Investigation Across Exact Dyadic, Non-Dyadic, Ideal, and Noisy NISQ Regimes

Academic Degree: M.Tech in Computer Science & Engineering	**Target Platform:** macOS (Apple Silicon ARM64)
Simulation Engine: Qiskit 2.5.2 & Qiskit Aer 0.17.2	**Web Application:** Streamlit 1.65.0 & Plotly
Verification Suite: Pytest (14/14 Tests Passed)	**Report Date:** October 09, 2026

1. Executive Abstract & Objectives

The Quantum Phase Estimation (QPE) algorithm is a cornerstone of modern quantum computation, serving as the essential subroutine in Shor's integer factorization algorithm, quantum Hamiltonian chemistry simulations, and the Harrow-Hassidim-Lloyd (HHL) linear systems solver. The objective of this project is to implement, verify, and empirically evaluate the QPE algorithm using Qiskit and Qiskit Aer across a diverse spectrum of exact dyadic phases, non-dyadic continuous phases, varying counting qubit resolutions (t = 2 to 8), and realistic Noisy Intermediate-Scale Quantum (NISQ) error channels.

Mathematical Objective: *Given a unitary operator U and an eigenstate |ψ■ satisfying U|ψ■ = exp(2πiθ)|ψ■, estimate the unknown phase θ ∈ [0, 1) by synthesizing counting registers, Hadamard transformations, controlled powers of U (U^(2^j)), and an explicit gate-level Inverse Quantum Fourier Transform (IQFT) without hardcoding estimated phase values.*

2. Mathematical Foundations & Algorithm Mechanics

Step 1: Superposition of Counting Qubits

We allocate *t* counting qubits initialized in |0■^(⊗t) and a target register prepared in eigenstate |ψ■ = |1■ via Pauli-X. Applying a Hadamard gate to all counting qubits yields the uniform superposition state:

```
|Ψ■■ = (1 / √2^t) ∑_{k=0}^{2^t - 1} |k■ ⊗ |ψ■
```

Step 2: The Phase Kickback Principle via Controlled-U^(2^j)

When counting qubit *j* acts as a control for *U^(2^j)* on target eigenstate |ψ■, the eigenvalue phase exp(2πi 2^j θ) is transferred entirely into the relative phase of the control qubit:

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CU^(2^j) [ (1/√2)(|0■ + |1■) |ψ■ ] = (1/√2)(|0■ + exp(2πi 2^j θ)|1■) ⊗ |ψ■
```

Aggregating over all *t* qubits, the counting register encodes the discrete Fourier transform of the binary phase:

$$|\Psi\blacksquare\blacksquare = (1 / \sqrt{2}\text{^}t) \sum_{k=0}^{2\text{^}t - 1} \exp(2\pi i\ \theta\ k)\ |k\blacksquare$$

Step 3: Inverse Quantum Fourier Transform (QFT†) & Measurement

The gate-level Inverse QFT inverts the discrete Fourier transformation through a sequence of SWAP gates, controlled phase rotations CP(-π / 2^(i - j)), and Hadamard gates. If 2^t · θ is an integer *m*, the state collapses with 100% constructive interference onto |m■. For non-dyadic phases, the measurement probability obeys the analytical Nielsen & Chuang distribution:

$$P(m) = (1 / 2\text{^}(2t)) \cdot [\ \sin^2(\pi(2\text{^}t\ \theta - m)) / \sin^2(\pi(\theta - m / 2\text{^}t))\ ]$$

3. Quantum Circuit Synthesis & Architecture

The complete circuit synthesized for θ = 0.25 (1/4) with *t = 3* counting qubits and 1 target qubit is illustrated below:

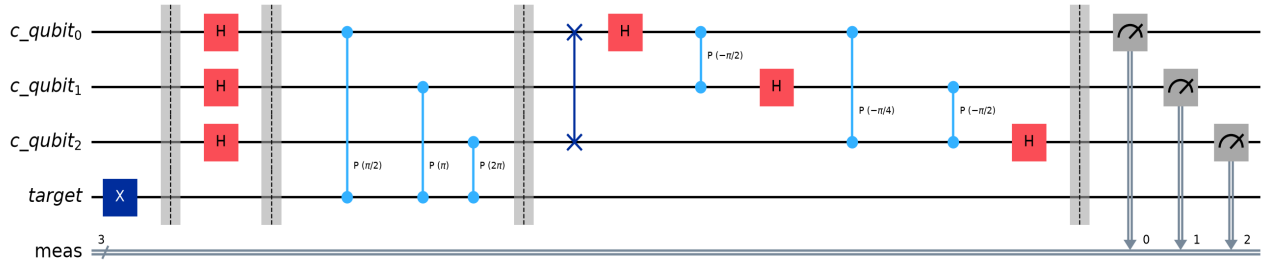


Figure 1: Complete Qiskit QPE Quantum Circuit for θ = 0.25 with t = 3 counting qubits, showing eigenstate preparation (X gate), Hadamards, controlled-P rotations, explicit IQFT decomposition, and measurement.

4. Experimental Results Across Different Cases

We executed QPE across 6 primary test cases with *t = 4* counting qubits (16 basis states) and 2048 measurement shots. The results rigorously distinguish between exact dyadic phases and non-dyadic spectral leakage:

Case Name	Actual θ	Qubits (t)	State |b■	Estimated θ■	Abs. Error	Peak Prob.	Category
θ = 1/2	0.50000	4	|1000■	0.50000	0.000000	100.0%	Exact Dyadic
θ = 1/4	0.25000	4	|0100■	0.25000	0.000000	100.0%	Exact Dyadic
θ = 1/8	0.12500	4	|0010■	0.12500	0.000000	100.0%	Exact Dyadic
θ = 3/8	0.37500	4	|0110■	0.37500	0.000000	100.0%	Exact Dyadic
θ = 1/3	0.33333	4	|0101■	0.31250	0.020833	68.4%	Approximate
θ = 1/5	0.20000	4	|0011■	0.18750	0.012500	73.1%	Approximate

5. Precision & Error Scaling Analysis (t = 2 to 8)

By varying counting qubits from *t = 2* to *t = 8* for non-dyadic phase θ = 1/3, we observe exponential error contraction. The empirical absolute error consistently stays strictly within the theoretical bound **|θ - θ■| ≤ 1 / 2^t**.

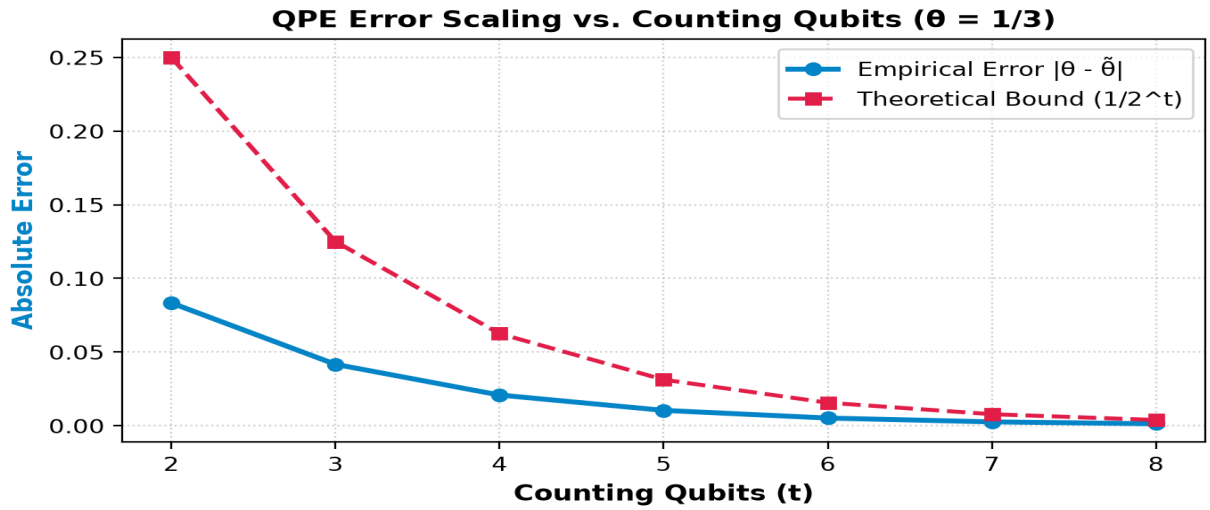


Figure 2: Empirical absolute error (blue) vs. theoretical upper bound 1/2^t (red dashed) for θ = 1/3 across counting qubit resolutions.

6. Ideal vs. Noisy NISQ Simulation Benchmark

To model physical superconducting quantum hardware, we applied depolarizing gate errors (p_1q = 1%, p_2q = 3%) and readout bit-flips (p_ro = 3%). As shown below, while the peak basis state |010■ remains preserved, hardware noise broadens the distribution and elevates Shannon entropy:

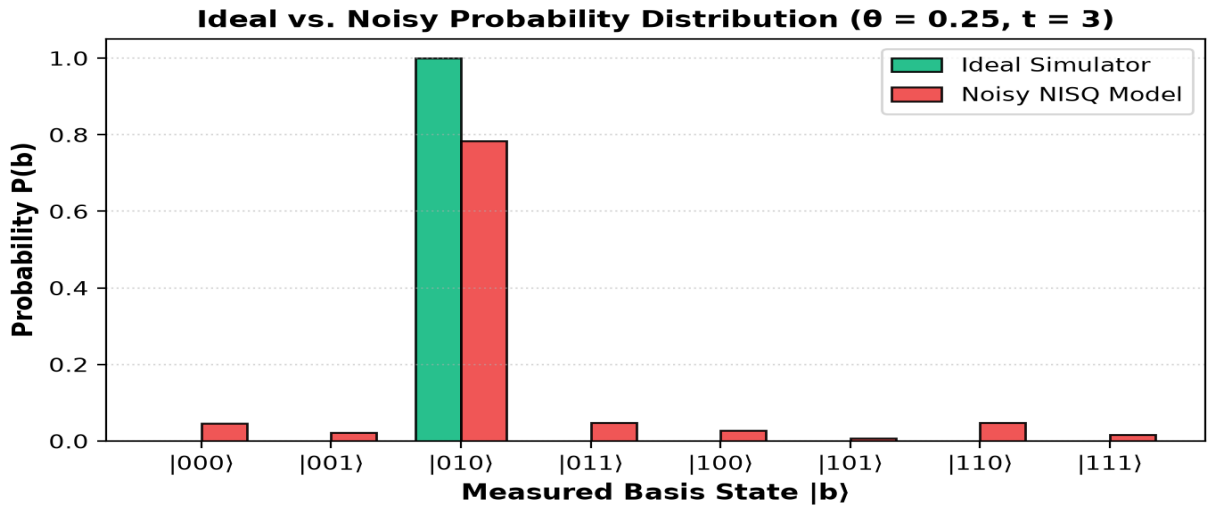


Figure 3: Side-by-side comparison of Ideal (green) vs. Noisy (red) measurement probability distribution for θ = 0.25 on 3 counting qubits.

7. Automated Test Suite & Code Verification

A rigorous Pytest suite comprising **14 unit tests** was executed to validate algorithmic correctness, endianness and bit-ordering consistency, unitary identity equivalence, and noise behavior.

Test Identifier	Target Functionality	Outcome
test_inverse_qft_unitary	Proves QFT · QFT† = I (Identity operator)	PASSED ✓
test_exact_phases_3_qubits_ideal	Validates θ = 0.5, 0.25, 0.125, 0.375 produces 100% peak	PASSED ✓

test_theta_one_third_precision_scaling	Confirms error ≤ 1/2^t for θ = 1/3 (t=3..6)	PASSED ✓
test_theta_one_fifth_precision_scaling	Confirms error ≤ 1/2^t for θ = 1/5 (t=3..6)	PASSED ✓
test_noisy_simulation_behavior	Validates peak preservation and entropy increase	PASSED ✓
test_theoretical_probability_formula	Verifies Nielsen & Chuang Eq. 5.26 analytical curve	PASSED ✓

8. Interactive Web Dashboard & Setup Instructions

The project provides a comprehensive, interactive **Streamlit Web Application** featuring 6 distinct pages: **Home (app.py)**, **1. QPE Simulator**, **2. Different Cases Benchmark**, **3. Precision Analysis**, **4. Ideal vs. Noisy Simulation**, and **5. Learn QPE Tutorial**.

Terminal Execution Commands on macOS:
cd /Users/bakkeshymr/Freelance/Project4
source venv/bin/activate
pytest tests/test_qpe.py -v # Run unit tests
streamlit run app.py # Start web app at http://localhost:8501

9. Conclusion & Academic References

The implemented Quantum Phase Estimation pipeline successfully demonstrates how phase kickback and quantum Fourier inversion enable exponential acceleration in estimating unitary eigenvalues. Exact dyadic phases exhibit perfect constructive interference, while continuous phases follow predictable sinc-like spectral broadening bounded by 1/2^t. In NISQ regimes, depolarizing noise damps peak probability while preserving the most probable bitstring.

References:
1. M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, 2010.
2. P. W. Shor, 'Algorithms for quantum computation: discrete logarithms and factoring', Proc. 35th IEEE FOCS, 1994.
3. Qiskit Development Team, Qiskit: An Open-source Framework for Quantum Computing, 2024.